

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. Name of the Academic Unit: Geology and Geophysics

2. Subject Name: Sedimentology

L-T-P: 3-0-0

Credits:3

3. Pre-requisites: None

4. Syllabus and reference books:

Syllabus:

Sedimentary particles and their sources; Physical processes and dynamics of moving particles including subaerial and sub-aqueous gravity flow/displacement processes; Size concepts, techniques of size analysis and application of size parameters for discriminating depositional environments; Particle morphology, factors that control particle morphology, Major categories of currents and the features in their deposits; sedimentary structures preserved on bed surfaces and undersurfaces of beds and their use for environmental interpretations, Biological and chemical processes and their products; Classification of sedimentary rocks i.e., classifying and naming intra-basinal and extra-basinal rocks; Heavy minerals and provenance studies; Diagenesis of sediments; Tectonics and sedimentation; Sedimentary environments i.e., marine and nonmarine, and use of their products for the recognition of sedimentary environments; Definition and classification of sedimentary basins; Lateral and vertical relationship and facies concept.

Reference Books:

- 1) Principles of Sedimentology and Stratigraphy (By Sam Boggs Jr.; Pearson; Latest edition)
- 2) Sedimentology: Process and Product (By M.R. Leeder, Springer; Latest edition)
- 3) Carbonate sedimentology (By Maurice E. Tucker, & V. Paul Wright; Wiley, Latest edition)
- 4) Depositional Sedimentary Environments (by H. E. Reineck and Indra Bir Singh; Springer-Verlag Berlin and Heidelberg GmbH & Co. Latest edition)
- 5) Sedimentology and Petroleum Geology (by Knut Bjorlykke, Springer)

5. Lecture-wise break-up:

Sl. No.	Topic	No. of lectures
1.	Why Sedimentology: An Introduction	2
2.	Sedimentary particles and their sources	2
3.	Physical processes and dynamics of moving particles including subaerial and sub-aqueous gravity flow/displacement processes	4
4.	Size concepts, techniques of size analysis and application of size parameters for discriminating depositional environments. Particle morphology,	3

	factors that control particle morphology.	
5	Major categories of currents and the features in their deposits; sedimentary structures preserved on bed surfaces and undersurfaces of beds and their use for environmental interpretations, Biological and chemical processes and their products	4
6	Biological and chemical processes and their products	2
7	Classification of sedimentary rocks i.e., classifying and naming intra-basinal and extra-basinal rocks; Heavy minerals and provenance studies	3
8	Diagenesis of sediments; Siliciclastic	2
9	Diagenesis of sediments-Carbonates	2
10	Tectonics and sedimentation	2
11	Sedimentary environments i.e., marine and nonmarine, and use of their products for the recognition of sedimentary environments	6
12	Definition and classification of sedimentary basins; Lateral and vertical relationship and facies concept	3
Total number of hours		35